

Product Backlog

Backlog ID	Epic/Story Title	User Story (if applicable)	Priority	Story Points	Acceptance Criteria Example	Sprint Target	Dependencies	Risks & Mitigations
EPIC-1	Familiarization with Existing Software	As a developer, I want to access and review the existing Python-based image-guided intervention software, so that I can understand its architecture and performance.	High	8	Given access to the existing software, When the system is run with provided data, Then the setup completes successfully and architecture components are documented.	Sprint 1	Access to existing repo and data	Missing dependencies → add environment setup scripts
STORY-1.1	Setup and Documentation	As a developer, I want to set up the existing software in a controlled environment, so that I can run and test it.	High	5	Given a configured environment, When setup scripts are executed, Then the software installs without errors and documentation describes setup steps clearly.	Sprint 1	EPIC-1	Environment conflicts → align dependencies across all setups
EPIC-2	Modularization and Extensibility	As a developer, I want the architecture to be modular and extensible, so that future enhancements can be efficiently integrated.	High	13	Given the existing architecture, When modules are refactored, Then the interfaces are clear and code dependencies are minimized.	Sprint 1	EPIC-1 complete	Integration bugs → introduce staged merges and validation
STORY-2.1	Refactor Modules	As a developer, I want to refactor the code into independent modules, so the system is easier to extend.	High	8	Given existing functions, When modules are separated, Then each compiles independently and retains the same functionality.	Sprint 1	EPIC-2	Modules don't connect properly → define clear API contracts

STORY-2.2	Tests	As a developer, I want to add unit and integration tests, so that the functionality is verified.	High	5	Given functional modules, When automated tests run, Then all pass successfully and full coverage.	Sprint 1	EPIC-2	Tests give inconsistent results → stabilize test data and automate regular test runs
EPIC-3	2D Images to 3D Volume Reconstruction	As a surgeon, I want accurate 3D reconstruction from 2D ultrasound images, so that I am able to make better surgical decisions.	High	13	Given a set of 2D ultrasound images, When the reconstruction pipeline runs, Then a 3D volume is produced within or improved reference accuracy.	Sprint 2	Modular architecture (EPIC-2)	Slow performance → use profiling to guide optimization
STORY-3.1	Analyze Current Code (Python and C++)	As a developer, I want to thoroughly analyze and review the existing reconstruction code, so that I can identify inefficiencies.	High	5	Given current reconstruction scripts, When profiling tools are executed, Then bottlenecks and memory inefficiencies are logged and documented.	Sprint 2	EPIC-3	Performance issues not obvious → use profiling to identify and focus on the slowest code sections
STORY-3.2	Prototype Optimization	As a developer, I want to prototype an optimized reconstruction pipeline, so that performance is improved.	High	8	Given identified bottlenecks, When optimization changes are applied, Then the system runs faster while maintaining at least equivalent 3D results.	Sprint 2	STORY-3.1	Optimizations could reduce accuracy → compare and validate outputs with baseline before merging